Com S 4720/5720 Fall 2025

Problem Set 5 (40 pts)

Due at 11:59pm

Friday, October 3

The first two problems are from *https://aimacode.github.io/aima-exercises/* (possibly) with minor typo fixes, slight rewriting, or addition of missing information.

1.(14 pts) (Exercise 5.9) This problem exercises the basic concepts of game playing, using tic-tac-toe (noughts and crosses) as an example. We define X_n as the number of rows, columns, or diagonals with exactly n X's and no O's. Similarly, O_n is the number of rows, columns, or diagonals with just n O's. The utility function assigns $+1$ to any position with $X_3 = 1$ and -1 to any position with $O_3 = 1$. All other terminal positions have utility 0. For nonterminal positions, we use a linear evaluation function defined as $\text{Eval}(s) = 3X_2(s) + X_1(s) - (3O_2(s) + O_1(s))$.

(a) (3 pts) Approximately how many possible games of tic-tac-toe are there?

(b) (4 pts) Show the whole game tree starting from an empty board down to depth 2 (i.e., one X and one O on the board), taking symmetry into account.

(c) (3 pts) Mark on your tree the evaluations of all the positions at depth 2.

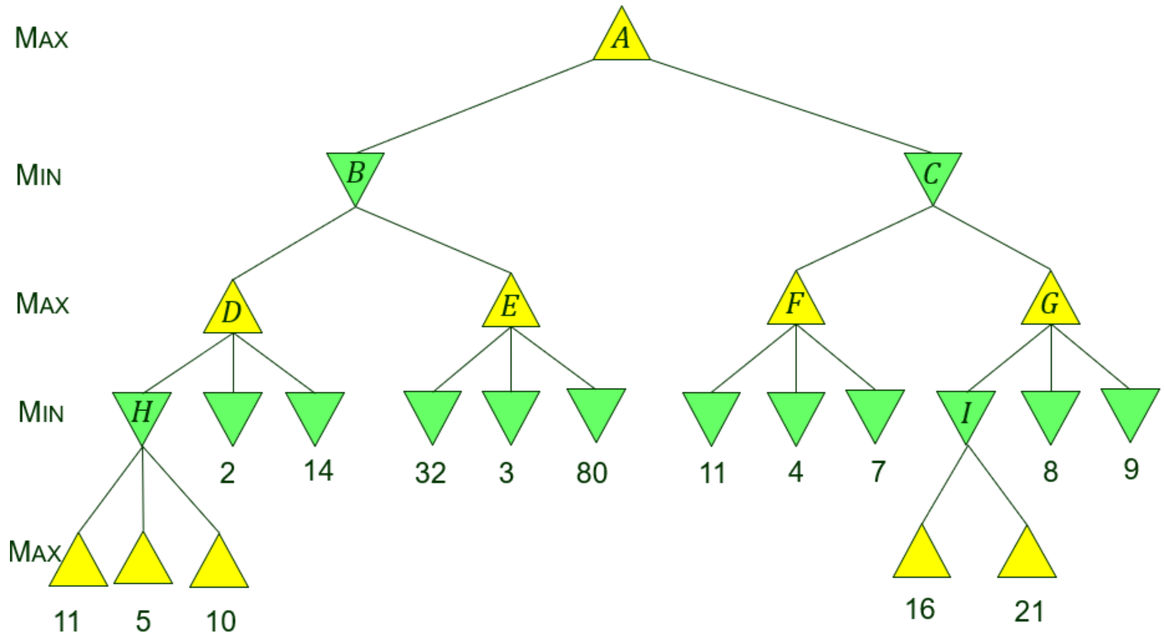
(d) (4 pts) Using the minimax algorithm, mark on your tree the backed-up values for the positions at depths 1 and 0, and use those values to choose the best starting move.

2.(20 pts) You are given a minimax search tree as shown on the next page. The tree has nine internal nodes $A, B, \ldots, I$. Not all terminal states (leaves) are at the same depth. Execute the alpha-beta pruning algorithm (use the version from the 3rd edition of the textbook in the lecture notes on September 25).

(a) (7 pts) Mark all the subtrees (including leaves) that have been pruned. You may, for instance, simply put double slashes `\\` or `//` across the edge entering the root of such a subtree from the above.

(b) (10 pts) Next to each visited internal node, write down the two values $[\alpha, \beta]$ *just before the return* from the call MAX-VALUE or MIN-VALUE invoked on the state represented by the node.

(c) (3 pts) What is the final value for MAX at the root?



3.(6 pts) (Exercise 6.1) How many solutions are there for the map-coloring problem in Figure 6.1 below? How many solutions if four colors are allowed? Two colors?

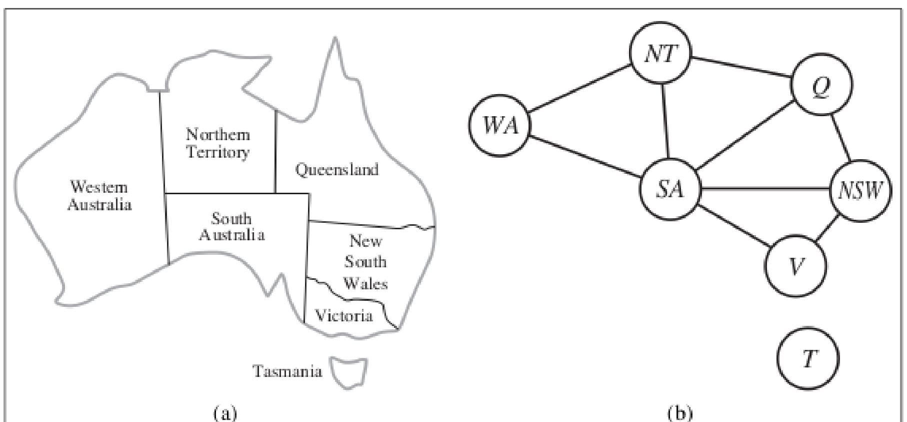


Figure 6.1 (a) The principal states and territories of Australia. Coloring this map can be viewed as a constraint satisfaction problem (CSP). The goal is to assign colors to each region so that no neighboring regions have the same color. (b) The map-coloring problem represented as a constraint graph.